

DELAWARE BAY BUOY 44009, USA

WMO: 992400

Lat: 38.457N

Lon: 74.702W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.3	-4.1	-14.5	1.1	-5.0	-12.5	1.3	-3.5	17.1	0.0	16.2	0.9	11.1	310	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.0	26.7	24.6	26.1	24.2	25.6	23.8	25.6	26.0	25.2	25.6	24.7	25.3	4.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.6	25.8	25.0	20.1	25.5	24.4	19.4	25.1	78.6	26.0	76.6	25.7	74.6	25.4	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	14.3	13.1	-8.0	28.5	2.5	1.0	-9.8	29.2	-11.3	29.8	-12.7	30.4	-14.5	31.2		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	3.7	3.5	5.8	9.9	14.6	20.2	23.3	23.5	21.3	16.7	11.8	7.3
	DBStd	7.76	4.30	3.46	2.84	2.40	2.66	2.48	1.55	1.59	2.04	2.91	3.47	4.02
	HDD10.0	663	196	181	131	30	1	0	0	0	0	1	23	101
	HDD18.3	2242	453	414	387	252	118	11	0	0	3	66	197	342
	CDD10.0	1953	2	0	2	28	144	306	412	419	339	207	76	17
	CDD18.3	492	0	0	0	0	3	67	154	162	92	14	0	0
	CDH23.3	1264	0	0	0	0	0	57	492	597	117	2	0	0
	CDH26.7	18	0	0	0	0	0	1	8	9	0	0	0	0
Wind	WSAvg	6.4	7.9	7.3	6.9	6.4	5.4	5.0	4.8	5.1	6.0	6.9	7.3	7.8
Precipitation	PrecAvg	1036	76	70	100	78	74	80	95	99	97	87	81	97
	PrecMax	1515	160	148	194	149	142	174	180	264	207	167	174	194
	PrecMin	796	24	13	3	31	14	36	37	13	3	1	6	38
	PrecStd	182	32	35	46	29	37	34	41	62	53	51	48	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	11.2	12.9	16.9	21.8	26.1	27.5	27.7	26.3	23.7	18.8	16.0
		MCWB	12.5	10.5	12.4	15.9	21.1	24.4	24.5	25.4	24.6	22.3	18.1	15.5
	2%	DB	11.7	9.9	11.5	15.0	20.4	24.7	26.7	26.8	25.3	22.1	17.7	15.0
		MCWB	11.3	9.2	11.1	14.5	19.7	23.6	24.5	24.7	23.9	21.0	17.0	14.4
	5%	DB	10.5	8.9	10.4	13.7	19.2	23.9	26.1	26.3	24.6	21.2	16.9	14.0
		MCWB	9.8	8.2	9.9	13.0	18.5	23.0	24.3	24.2	22.9	19.8	15.8	13.4
	10%	DB	9.3	7.9	9.4	12.9	18.1	23.2	25.5	25.7	24.0	20.4	16.1	12.9
		MCWB	8.5	7.1	8.7	11.9	17.5	22.3	24.0	23.7	22.0	18.7	14.9	12.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	11.2	11.8	16.4	20.6	25.2	26.1	27.1	25.6	22.9	18.5	16.0
		MCDB	13.1	11.4	12.0	17.2	20.7	25.6	26.3	27.4	25.8	23.4	18.6	16.0
	2%	WB	11.6	9.8	10.6	14.9	19.4	24.3	25.5	25.7	25.0	21.8	17.5	14.8
		MCDB	11.8	10.1	10.8	15.2	19.6	24.7	25.8	26.2	25.4	22.4	17.8	14.9
	5%	WB	9.9	8.5	9.6	13.3	18.6	23.6	25.1	25.2	23.9	20.8	16.4	13.7
		MCDB	10.2	8.8	9.8	13.6	18.9	23.9	25.4	25.9	24.7	21.5	16.8	14.0
	10%	WB	8.5	7.3	8.4	12.0	17.8	22.9	24.6	24.7	22.9	19.8	15.5	12.4
		MCDB	9.0	7.7	8.8	12.3	18.1	23.2	25.1	25.5	23.7	20.5	16.0	12.8
Mean Daily Temperature Range	5% DB	MDBR	4.3	4.0	3.4	3.0	2.6	2.4	2.3	2.0	2.1	2.9	3.4	3.9
		MCDBR	4.9	4.6	3.9	3.6	3.2	3.1	2.6	2.5	2.1	3.0	3.5	4.2
		MCWBR	5.8	5.4	4.4	3.3	3.0	2.9	2.6	2.7	2.8	3.9	4.4	5.6
	5% WB	MCDBR	4.7	4.7	4.0	3.8	2.8	2.7	2.0	2.2	1.9	2.7	3.1	4.3
		MCWBR	5.7	5.5	4.5	3.6	2.9	2.8	2.2	2.4	2.6	3.8	4.2	5.6
Clear-Sky Solar Irradiance	taub	0.322	0.327	0.357	0.393	0.440	0.484	0.500	0.498	0.413	0.366	0.334	0.322	
	taud	2.512	2.496	2.443	2.357	2.261	2.184	2.152	2.146	2.371	2.497	2.533	2.541	
	Ebn at Noon	859	902	903	885	845	806	788	780	834	846	837	832	
	Edh at Noon	78	90	104	121	135	146	150	148	111	89	76	71	
All-Sky Solar Radiation	RadAvg	2.08	2.92	4.05	5.18	5.88	6.42	6.26	5.61	4.58	3.41	2.39	1.83	
	RadStd	0.19	0.29	0.34	0.38	0.63	0.53	0.33	0.40	0.44	0.32	0.23	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+79

Nomenclature: See separate page